

Japan is showing us the way when it comes to acceptance of robots in our daily lives. In that country, they're not viewed as potential monstrosities: rather, they're gentle, benign creatures. One reason cited by some is the Japanese (Shinto) religion, where every creature—including robots—have a "life force." There are accounts of how robotic pets reduce stress and make life nicer for old folks whose children don't visit them often—in one old-age home, two robotic seals function as pets for the residents, who are delighted to stroke them and feel them flapping their flippers. "Old fogies," you might say. Or, "Why not just give them a dog?" We're wondering about the dog too, but the point is that the Japanese really seem to more readily assign a "human" status to "dumb, mechanical creatures."

The point also is, if it can happen in Japan, it can't just be explained away by Shintoism; it will happen everywhere, sooner or later. And not just robotic seals and dogs: as the whole humanoid thing catches on, who's to say that old (or young) people, for instance, won't accept robots as "real"?

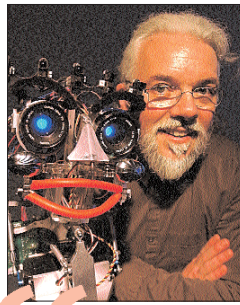
What They're Saying

Hans Moravec, old-time researcher at Carnegie-Mellon University: "The effect of robots clearly has implications for the economy. Social security will have to be expanded, introduced at lower and lower ages, till essentially everyone lives on social security. The taxes will be paid by fully-automated businesses run by robots. And human beings will have to deal with the problem of excess leisure." (What would you do when there's nothing to do?)

Ron Arkin, roboticist at the Robotics and Artificial Intelligence Group at LAAS/CNRS in France: "I would expect people to form psychological attachments with these artefacts. We already do so more or less: to cars to video games to AIBOs. ...Would you want your daughter to marry a robot? What would the church think of such unions?" (This one might sound ridiculous, but think about *your* psychological attachments to artefacts.)

Daniela Cerqui, social and cultural anthropologist at the Institute of Sociology and Anthropology of the University of Lausanne: "I am afraid that the long term future we are building will have no space left for human beings." (To do anything useful, we presume.)

We're not saying the views above are indicative of Truth. But they highlight what the thinkers



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Noel Sharkey

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are thinking right now: that it's a foregone conclusion that the bots will be amongst us—in growing numbers—now on.

What's To Come?

Honestly, we haven't the foggiest. But let's take pot-shots at possibilities.

What gets some people's brows all sweaty is questions like "What if they became more intelligent than us?" and "Will they take over?" (That, by the way, has been asked by people about 16 billion times, which is 2.46 times per person)

Turns out, in 2001, someone of no less stature than that of Stephen Hawking recommended that humans change their DNA through genetic modification to stop intelligent machines from "taking over the world." What can we say?

Not much, but Prof Noel Sharkey has license to say something: that contrary to popular belief, robots have no chance of taking over the world. "They don't have as much intelligence as bacteria and they never will." Sharkey is the man behind Robot Wars (see *Robo Sapiens, Digit*, January 2004), where evolutionary robots were divided into prey and predators, and fought it out. Would Sharkey and Hawking please fight it out and tell us something we can believe!

Perhaps we *should* believe someone with a name like Marshall Brain. He happens to be the founder of howstuffworks.com, and at the much-read <http://marshallbrain.com/robotic-nation.htm>, Brain envisions a situation where robots will have all the jobs.

Actually, all talk of "robots on the rise" is complementary to talk about cyborgs and how we will turn cyborg—there's the possibility of a mixed race on the planet, each individual of either machine or human origin, and with sharply varying degrees of intelligence and physical capabilities. Lending credence to this idea is Ray Kurzweil, inventor and futurist: "By the time we succeed in building (intelligent) machines, we will have become part machine ourselves."

Kevin Warwick, the world's first self-proclaimed cyborg, who wrote *Human-Machine Convergence* for *Digit* in June 2005, says about the people who don't "upgrade": "If they don't, they could become some sort of subspecies."

The "rise of the robots" doesn't need any spectacular advances in AI, for two reasons: first, AI as it stands today aims at imparting human-like intelligence to machines, and the new machines may not need such intelligence—they could well develop a different kind. And second, even if you do take the stance that our machines are to mimic us, the sheer numbers overwhelm. Take common sense, for example, which is notoriously difficult for a machine to have: with computational power increasing exponentially—and the possibility of quantum computers (see *The Quest For Quantumland, Digit*, November 2006)—common sense could be hard-coded into artificial brains by brute-force methods, without our even having to understand our own common sense!

As you let your imagination loose and try and imagine the future—which is all we can encourage you to do—try and remember that humanness is overrated. It's gotten down to that. ■

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They're Crazy!

According to a recent study by the British government, robots could one day demand the same "citizen's rights" as humans. If granted, countries would be obliged to provide social benefits—including housing. Don't laugh at us—that's what the officiously-called "study" says!

A "monumental shift" could occur if robots develop to the point where they can reproduce, improve themselves or develop artificial intelligence; that at some point in the next 20 to 50 years, robots could be granted rights. If this happened, the bots would have responsibilities such as voting, the paying of taxes, and serving in the military.

These predictions are contained in papers that look at projected developments over the next 50 years.

If you're thinking we made this up, take a look at <http://news.bbc.co.uk/2/hi/technology/6200005.stm>.